



8921 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evolution?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam Imaging				
	1		NIRCam Imaging	(1) NGVSUDG-05
	2		NIRCam Imaging	(2) NGVSUDG-19

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	3		NIRCam Imaging	(3) NGVSUDG-20
NIRSpec/MOS				
	7	NGVSUDG-05	NIRSpec MultiObject Spectroscopy	(4) NGVSUDG-05_ngvscat
	8	NGVSUDG-19	NIRSpec MultiObject Spectroscopy	(5) NGVSUDG-19_ngvscat
	9	NGVSUDG-20	NIRSpec MultiObject Spectroscopy	(6) NGVSUDG-20_ngvscat

ABSTRACT

Ultra-diffuse galaxies (UDGs) show a puzzling diversity in their properties, from their globular cluster (GC) content to their dark matter content. Only 27 UDGs to date have measured kinematics (mostly from spectroscopy of GCs), and these data have shown that UDGs have velocity dispersions that span a remarkable range, ~ 10 -100 km/s, in a narrow range of stellar mass. While several formation mechanisms exist to explain the formation of UDGs in dwarf-mass halos with velocity dispersions < 30 km/s, UDGs with dispersions > 50 km/s may only be explained by a dark matter distribution consistent with Milky-Way like halos. Such 'failed galaxies' are missing in all current cosmological simulations. A recent study has identified three UDGs in the Virgo cluster that are extreme outliers, with GC velocity dispersions suggesting virial masses of $\sim 10^{12}$ Msun. These estimates are based on 3-6 GCs per galaxy and need confirmation with larger numbers of GCs that only JWST can provide. We propose to use the NIRSpec MSA to go 1.5 mag deeper than is possible from the ground, tripling the number of GC velocities in these UDGs, and determining whether they are indeed the most massive failures in the Local Universe.

OBSERVING DESCRIPTION

We propose to use the NIRSpec MSA to obtain radial velocities for globular clusters in three Virgo cluster ultra-diffuse galaxies. We will use the G235H grating with the F170LP filter to target the 1.7-3.0 micron spectral region, which contains numerous sharp features (H, CO) with which to measure radial velocities. Because there is no extant imaging with sufficiently precise astrometry, we will obtain pre-imaging using NIRCam with the F150W and F356W filters in the 8NIRSPEC dithering pattern. We will use the NIRCam imaging to select GC candidates using both color and size.

Proposal 8921 - Targets - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evolution?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGVSUDG-05	RA: 12 25 37.6098 (186.4067075d) Dec: +10 14 58.62 (10.24962d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Galaxy Description=[Dwarf elliptical galaxies, Low Surface Brightness galaxies] Extended=YES				
	(2)	NGVSUDG-19	RA: 12 33 29.5900 (188.3732917d) Dec: +15 14 2.80 (15.23411d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Galaxy Description=[Dwarf elliptical galaxies, Low Surface Brightness galaxies] Extended=YES				
	(3)	NGVSUDG-20	RA: 12 35 12.8600 (188.8035833d) Dec: +07 03 22.40 (7.05622d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Galaxy Description=[Dwarf elliptical galaxies, Low Surface Brightness galaxies] Extended=YES				
	(4)	NGVSUDG-05_ngvscat	RA: 12 25 37.2867 (186.4053612d) Dec: +10 14 36.46 (10.24346d) Equinox: J2000		
	Comments: Description=[]				
	(5)	NGVSUDG-19_ngvscat	RA: 12 33 32.2641 (188.3844337d) Dec: +15 14 10.17 (15.23616d) Equinox: J2000		
	Comments: Description=[]				
	(6)	NGVSUDG-20_ngvscat	RA: 12 35 13.4581 (188.8060754d) Dec: +07 03 34.52 (7.05959d) Equinox: J2000		
	Comments: Description=[]				

Proposal 8921 - Observation 1 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Observation	Proposal 8921, Observation 1										Thu Nov 20 17:00:29 GMT 2025								
	Diagnostic Status: Warning																		
	Observing Template: NIRCcam Imaging																		
	Comments: APA requirements to avoid claws artifacts from being >20% of sky background																		
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.																		
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.																		
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.																		
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.																		
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous									
	(1)	NGVSUDG-05	RA: 12 25 37.6098 (186.4067075d)				Epoch of Position: 2000												
			Dec: +10 14 58.62 (10.24962d)																
			Equinox: J2000																
Template	Comments:																		
	Category=Galaxy																		
	Description=[Dwarf elliptical galaxies, Low Surface Brightness galaxies]																		
	Extended=YES																		
Module	Subarray				Target Placement														
	ALL				FULL				Module Gap										
#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions										
	1		FULLBOX		8NIRSPEC		SMALL-GRID-DITHER		3										
#	Short Filter		Long Filter		Readout Pattern		Groups/Int		Integrations/Exp		Total Integrations		Total Dithers		Total Exposure Time		Optional ETC ID		
	1		F150W		F356W		BRIGHT1		4		1		24		24		1803.777		253560
Special Requirements	Sequence Visits within 53.0 Days																		
	Aperture PA Range 0 to 99 Degrees (V3 0.07457694 to 99.07457694)																		
Aperture PA Range 102 to 102 Degrees (V3 102.07457694 to 102.07457694)																			
Aperture PA Range 115 to 115 Degrees (V3 115.07457694 to 115.07457694)																			
Aperture PA Range 117 to 117 Degrees (V3 117.07457694 to 117.07457694)																			
Aperture PA Range 144 to 236 Degrees (V3 144.07457694 to 236.07457694)																			
Aperture PA Range 246 to 254 Degrees (V3 246.07457694 to 254.07457694)																			
Aperture PA Range 262 to 359 Degrees (V3 262.07457694 to 359.07457694)																			
Visits Same PA																			
7 After 1 by 60.0 Days to <None specified>																			

Proposal 8921 - Observation 2 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Observation	Proposal 8921, Observation 2										Thu Nov 20 17:00:29 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
	Comments: APA requirements to avoid claws artifacts from being >10% of sky background										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(2)	NGVSUDG-19	RA: 12 33 29.5900 (188.3732917d)				Epoch of Position: 2000				
			Dec: +15 14 2.80 (15.23411d)								
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Dwarf elliptical galaxies, Low Surface Brightness galaxies] Extended=YES										
Template	Module		Subarray					Target Placement			
	ALL		FULL					Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		8NIRSPEC		SMALL-GRID-DITHER				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F356W	BRIGHT1	4	1	24	24	1803.777		
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 0 to 31 Degrees (V3 0.07457694 to 31.07457694) Aperture PA Range 38 to 48 Degrees (V3 38.07457694 to 48.07457694) Aperture PA Range 63 to 96 Degrees (V3 63.07457694 to 96.07457694) Aperture PA Range 107 to 109 Degrees (V3 107.07457694 to 109.07457694) Aperture PA Range 114 to 116 Degrees (V3 114.07457694 to 116.07457694) Aperture PA Range 120 to 188 Degrees (V3 120.07457694 to 188.07457694) Aperture PA Range 198 to 359 Degrees (V3 198.07457694 to 359.07457694) Visits Same PA										
	8 After 2 by 60.0 Days to <None specified>										

Proposal 8921 - Observation 3 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Observation	Proposal 8921, Observation 3										Thu Nov 20 17:00:29 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
	Comments: APA requirements to avoid claws artifacts from being >10% of sky background										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(3)	NGVSUDG-20	RA: 12 35 12.8600 (188.8035833d)				Epoch of Position: 2000				
			Dec: +07 03 22.40 (7.05622d)								
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Dwarf elliptical galaxies, Low Surface Brightness galaxies] Extended=YES										
Template	Module		Subarray					Target Placement			
	ALL		FULL					Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		8NIRSPEC		SMALL-GRID-DITHER				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F356W	BRIGHT1	4	1	24	24	1803.777		
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 0 to 4 Degrees (V3 0.07457694 to 4.07457694) Aperture PA Range 12 to 50 Degrees (V3 12.07457694 to 50.07457694) Aperture PA Range 60 to 69 Degrees (V3 60.07457694 to 69.07457694) Aperture PA Range 75 to 78 Degrees (V3 75.07457694 to 78.07457694) Aperture PA Range 112 to 241 Degrees (V3 112.07457694 to 241.07457694) Aperture PA Range 243 to 351 Degrees (V3 243.07457694 to 351.07457694) Aperture PA Range 358 to 359 Degrees (V3 358.07457694 to 359.07457694) Visits Same PA										
	9 After 3 by 50.0 Days to <None specified>										

Proposal 8921 - Observation 7 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Observation	Proposal 8921, Observation 7: NGVSUDG-05											Thu Nov 20 17:00:29 GMT 2025
	Diagnostic Status: Error											
Observing Template: NIRSpec MultiObject Spectroscopy												
Diagnostics	(NGVSUDG-05 (Obs 7)) Error (Form): This observation was created with an Aperture PA of 270.0000 but it has been assigned an Aperture PA of 67.8549											
	(Aperture PA) Error (Form): This observation was created with an Aperture PA of 270.0000 but it has been assigned an Aperture PA of 67.8549											
	(Visit 7:1) Error (Form): Reference stars are required but none were found for this visit											
	(NGVSUDG-05 (Obs 7)) Warning (Form): Config c1 (#1) has 4 master background shutters affected by failed open or closed shutters.											
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	(Visit 7:1) Warning (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
Fixed Targets	(4)	NGVSUDG-05_ngvscat	RA: 12 25 37.2867 (186.4053612d)									
			Dec: +10 14 36.46 (10.24346d)									
Acquisition			Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	false		No	MSA Center	22 < g < 25 (1303 sources)		jwst-nirspec-g235h		1.5		
Reference Stars												
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (G235H/F170LP)	c1	3 Shutter Slitlet	186.41072070833 334 Degrees 10.264811111111 111 Degrees	67.855929640898 81			3	9	13261.301	

Proposal 8921 - Observation 7 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Special Requirements	MSA Scheduled Aperture PA 67.8549 to 67.8549 Degrees (V3 289.28036 to 289.28036) 7 After 1 by 60.0 Days to <None specified>
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Proposal 8921 - Observation 8 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Observation	Proposal 8921, Observation 8: NGVSUDG-19											Thu Nov 20 17:00:29 GMT 2025
	Diagnostic Status: Error											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(NGVSUDG-19 (Obs 8)) Error (Form): This observation was created with an Aperture PA of 270.0000 but it has been assigned an Aperture PA of 69.1610											
	(Aperture PA) Error (Form): This observation was created with an Aperture PA of 270.0000 but it has been assigned an Aperture PA of 69.1610											
	(Visit 8:1) Error (Form): Reference stars are required but none were found for this visit											
	(NGVSUDG-19 (Obs 8)) Warning (Form): Config c1 (#1) has 1 master background shutters affected by failed open or closed shutters.											
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	(Visit 8:1) Warning (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
	#	Name		Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(5)	NGVSUDG-19_ngvscat		RA: 12 33 32.2641 (188.3844337d) Dec: +15 14 10.17 (15.23616d) Equinox: J2000								
Acquisition	Comments: Description=[]											
	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method		HFF Readout Mode		Obtain Confirmation Images	Science Aperture		Primary Candidate List		Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
	MSATA		false		No	MSA Center		22 < g <25 (1232 sources)			jwst-nirspec-g235h	1.5
Reference Stars												
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (G235H/F170LP)	c1	3 Shutter Slitlet	188.37335875000 002 Degrees 15.234086111111 111 Degrees	69.158047096466 42			3	9	13261.301	

Proposal 8921 - Observation 8 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Special Requirements	MSA Scheduled Aperture PA 69.1610 to 69.1610 Degrees (V3 290.5864 to 290.5864) 8 After 2 by 60.0 Days to <None specified>
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Proposal 8921 - Observation 9 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Observation	Proposal 8921, Observation 9: NGVSUDG-20											Thu Nov 20 17:00:29 GMT 2025
	Diagnostic Status: Error											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(NGVSUDG-20 (Obs 9)) Error (Form): This observation was created with an Aperture PA of 260.0000 but it has been assigned an Aperture PA of 68.3130											
	(Aperture PA) Error (Form): This observation was created with an Aperture PA of 260.0000 but it has been assigned an Aperture PA of 68.3130											
	(Visit 9:1) Error (Form): Reference stars are required but none were found for this visit											
	(NGVSUDG-20 (Obs 9)) Warning (Form): Config c1 (#1) has 1 master background shutters affected by failed open or closed shutters.											
	(NGVSUDG-20 (Obs 9)) Warning (Form): We recommend you request at least 60 days of lead time (via the After Observation Link) in order to reduce the Pre-Image observation.											
	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	(Visit 9:1) Warning (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
	#	Name		Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(6)	NGVSUDG-20_ngvscat		RA: 12 35 13.4581 (188.8060754d) Dec: +07 03 34.52 (7.05959d) Equinox: J2000								
Acquisition	Comments: Description=[]											
	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
Template	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
Reference Stars	MSATA	false		No	MSA Center	NGVSUDG-20 22<g<25 (1164 sources)		jwst-nirspec-g235h		1.5		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (G235H/F170LP)	c1	3 Shutter Slitlet	188.79531608333 332 Degrees 7.0549475 Degrees	68.311690657058 36			3	9	13261.301	

Proposal 8921 - Observation 9 - Learning From Your Failures: Are these Ultra-Diffuse Galaxies the Massive Failures of Galaxy Evoluti...

Special Requirements	MSA Scheduled Aperture PA 68.3130 to 68.3130 Degrees (V3 289.73846 to 289.73846) 9 After 3 by 50.0 Days to <None specified>
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